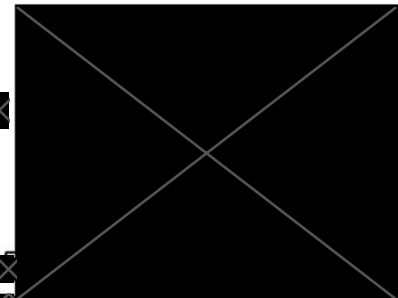


Professor [REDACTED]
Professor Emeritus

[REDACTED]
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8

11 September 2002

Re: Evaluation of energy devices.

J [REDACTED] Many of these persons identify a "over unity machine" as a perpetual motion device. Nothing could be more from the truth. In truth it should be stated as the "obvious energy over unity machine". In fact if the ZPE energy could be accurately measured, the total output would be less than the total input. In fact those who I think have accomplished a ZPE tap were all concerned that the total local ZPE is less than initially and may have consequences. So in fact there are NO perpetual motion machines, nor machines with total energy out greater total energy in. It is only what appears to be the easily observable energy out compared to the easily observable energy out is greater than unity.

1. [REDACTED] Aurora, Colorado, Magnetic Piston Reciprocating Engine. To move objects, accelerate objects to speed requires energy, significant energy. A representative space launcher requires about 5 billion Btu's in fuel energy to reach orbital speed and altitude. About 55% of that energy is converted to kinetic and potential energy. A series, of what appear to be, permanent magnet pistons moving in a cylinder have no such energy potential. In fact a 14 gallon gasoline tank on an automobile contains about 39,000 Btu's. If this device were built it is not likely that it would function as described. In fact the last sketch in Mr. Wing's submission, the airplane, is a convincing negative statement.

2. [REDACTED] Council Bluffs, Iowa. Dual Rotor Alternator. Although Mr. McDonald submitted an awarded patent, it is unclear as to how this can be used as a power device. The energy to rotate the alternator is a function of the electrical energy generated. The internal friction is real, but not significant compared to the energy generated. An alternator is not configured to be an AC motor, even if it were the alternator output to the AC motor would be less than the energy in the wheel. So if the wheel were at speed, it could not maintain that speed because the net energy to the AC motor MUST always be less than that developed by the wheel. His item one has the flavor of the Meyer brothers, Jeffery Gross and Ken Shoulders. Item 2 is a question mark for a wet sump hydrogen engine. Item three has been around along time and discarded for a long time. Item four is in existence and is based on the original Russian work on ultra-short pulse (10 pico seconds) ultra wide band radar, and that he does not mention. Item 5 is the SERJ (Supercharged Ejector Ram/Scram Jet) built and tested to Mach 8 in 1966. It delivered 5% better performance than predicted and was summarily canceled after its first successful run! It was attempted to be re-born last year and the same type people that killed it in 1966 killed it in 2001. Cost per pound to orbit is almost solely dependent on fleet flight rate. If three shuttles could fly at 90 flight per year not 9, the cost per pound would be \$2,000 per

pound not \$57,000 per pound. Item 6 is interesting but more available than he might be aware. His last claim is possibly his best claim, and the alternator may not be the key item.

3. [REDACTED] Canada. **Bootstrap Motor-Generators**. No matter what the stored energy in the flywheels, the power output is limited to the total energy divided the output time. Unless there is some ZPE or other input you cannot amplify a small input into a 5000 Watt output on a continuous basis. The bicycle experiment showed energy can be stored, but it cannot be increased beyond what is stored.

4. [REDACTED] Holland Michigan. **Transduction Device**. I am afraid the answer to the question of input to output energy seals the fate of this discussion. It is political physics and a fake perpetual motion machine. Some of his sketches resemble space tethers, but even they have an input to output ratio of less than one. This is gobbly-gook of the first magnitude.

M [REDACTED] y, Location unknown. **Closed Loop Implosion Turbine**. The three phenomena he mentions, laminar flow, vortices, and cavitation all have energy input to output ratios of less than one. In fact the latter two are great energy dissipators that destroy available energy. Whenever I see things like "Repusine" instead of water and generating a piezo field then the statement that the recipient need not be caught up on or persuaded to ZPE and other subtle energies there is something wrong! This sounds like pseudo-science. The Tokamak is a failure to date and his explanation is wild.

I think Bearden has more credibility.

[REDACTED]